

FRIEDRICH SCHILLER UNIVERSITY JENA

FACULTY OF ECONOMICS AND BUSINESS ADMINISTRATION

Securitisation Markets and Bank Lending: Evidence from European Banks

MASTER THESIS SUBMITTED IN PARTIAL FULFILLMENT OF THE
REQUIREMENTS FOR THE DEGREE OF MASTER OF SCIENCE (M.SC) IN
ECONOMICS

ISSABELLA SONNE

STUDENT ID: 208550

Supervisor:

Prof. Daniel Streitz

Chair of Finance, Law, and Regulation

Faculty of Economics and Business Administration

Friedrich Schiller University Jena

Contents

1. Introduction	1
2. Literature Review	4
3. Data and Main Variables	6
3.1. Data Sources and Sample Construction	6
3.1.1. Bank Lending Measures	7
3.1.2. Bank-Level Control Variables	8
3.1.3. Country-Level Securitisation Activity	9
3.2. Summary Statistics	11
3.3. Descriptive Evidence on Securitisation and Lending in the Eurozone . .	12
4. Securitisation Market Depth and Bank Lending	16
5. Regression Analysis	17
5.1. Econometric Model	18
5.1.1. Baseline Results	19
5.1.2. Robustness Checks: Loan Coverage and Reporting Consistency	22
5.1.3. Robustness by Loan Type: Corporate versus Non-Corporate Lending	23
5.2. Heterogeneity Across Banks	26
5.2.1. Heterogeneity by Bank Capital Strength	27
5.2.2. Heterogeneity by Bank Risk	27
5.3. Country-Level Evidence	29
6. Discussion and Conclusion	32
A. Appendix	36
B. Declaration of Academic Integrity	45

1. Introduction

Securitisation plays an important role in modern financial systems by allowing banks to transform illiquid loans into tradable securities. By converting loans into marketable claims, banks can expand their lending capacity without proportionally increasing risk-weighted assets, particularly when credit risk is effectively transferred to external investors. This mechanism relaxes funding and regulatory capital constraints and may thereby support credit provision to the real economy. For this reason, securitisation has long been regarded as a major financial innovation in banking.

Early theoretical contributions emphasise the role of securitisation in improving risk allocation and mitigating informational frictions. G. B. Gorton and Pennacchi (1995) show that loan sales and tranching can reduce asymmetric information problems between originators and investors by structuring claims in a way that preserves incentives. Similarly, Duffie and Gârleanu (2001) develop a model of credit risk transfer markets in which securitisation reallocates risk toward investors with greater risk-bearing capacity. In these frameworks, securitisation enhances liquidity and contributes to more complete financial markets.

At the same time, securitisation influences bank behaviour through balance sheet channels. By removing loans from the balance sheet, banks may reduce regulatory capital requirements and free up capacity for new lending. It can also improve liquidity management and mitigate maturity mismatches, particularly for institutions facing binding capital or funding constraints. These mechanisms suggest that securitisation may affect not only the overall level of lending but also its composition across borrower groups.

Empirical evidence from the United States largely supports the view that securitisation affects credit supply. Loutskina and Strahan (2009) show that banks with greater access to securitisation markets are less sensitive to local deposit supply shocks, consistent with securitisation providing an alternative funding channel. Mian and Sufi (2009) document that the expansion of securitised mortgage credit prior to the Global Financial Crisis was associated with increased lending to lower-credit-quality borrowers. At the same time, the crisis revealed important distortions. Keys et al. (2010) provide evidence consistent with weakened screening incentives around securitisation eligibility thresholds, while Acharya et al. (2013) show that securitisation was sometimes used for regulatory capital arbitrage, with limited effective risk transfer. Taken together, this literature suggests that securitisation can expand credit supply but may also generate incentive problems and risk-shifting behaviour.

In Europe, the development of securitisation markets followed a different trajectory. After the Global Financial Crisis, European securitisation activity remained persistently subdued, in contrast to the relatively rapid recovery observed in the United States. This divergence attracted considerable policy attention. The European Central Bank and the European Commission have repeatedly argued that revitalising securitisation markets, particularly through simple, transparent and standardised (STS)¹ instruments, could enhance risk sharing and support lending in bank-dependent financial systems (European Central Bank, 2018; European Commission, 2015). Because firms in the euro area rely heavily on bank financing, changes in banks' funding conditions may have particularly strong macroeconomic implications.

Despite its policy relevance, empirical evidence on securitisation and bank lending in Europe remains mixed. Pre-crisis studies such as Altunbas et al. (2009) suggest that securitisation relaxed balance sheet constraints for European banks and weakened the traditional bank lending channel of monetary policy. Subsequent research indicates that lending effects depend on bank characteristics, regulatory conditions, and market environments (Benetton et al., 2021; Casu et al., 2013). Moreover, much of the existing literature relies on bank-level securitisation measures derived from balance sheet disclosures, which often suffer from limited coverage and reporting inconsistencies.

In this thesis, I adopt a complementary approach. Rather than focusing on bank-level issuance, I examine country-level securitisation market depth. European securitisation markets operate within national legal frameworks, supervisory regimes, and investor bases that evolve gradually over time and are largely common to banks operating within the same jurisdiction. Variation in securitisation activity at the country level therefore reflects changes in the broader institutional and market environment rather than bank-specific reporting decisions by individual banks. Exploiting this variation allows the analysis to examine whether changes in securitisation market depth are systematically associated with bank lending outcomes.

Accordingly, this thesis addresses the following research question: *Does securitisation market depth influence the level and composition of bank lending in the post-crisis euro area?*

To approach this question, I combine country-level securitisation data from the ECB's

¹The STS securitisation framework was established by Regulation (EU) 2017/2402 and came into application in 2019 as part of the Capital Markets Union framework. See the European Commission for an overview: https://finance.ec.europa.eu/securitisation_en.

Financial Vehicle Corporation statistics with bank-level balance sheet information from BankFocus covering euro-area banks over the period 2009–2022. Securitisation activity is measured as the outstanding stock of securitised assets relative to GDP. The empirical analysis exploits variation across countries and over time while controlling for bank characteristics and common macroeconomic developments.

The analysis further investigates whether the relationship between securitisation and lending varies across loan types and bank characteristics. In particular, I distinguish between corporate and non-corporate lending and examine heterogeneity by capitalisation and credit risk. These extensions are motivated by theoretical models suggesting that balance sheet relief mechanisms may be especially relevant for banks facing binding capital or liquidity constraints (Kashyap et al., 2002; Pennacchi, 1988).

This thesis contributes to the literature in three ways. First, it provides new empirical evidence on the relationship between securitisation and bank lending in the post-crisis euro area. Second, it introduces a country-level measure of securitisation market depth that captures institutional variation across jurisdictions. Third, it examines heterogeneity across loan types and bank characteristics, shedding light on potential compositional effects of securitisation on credit supply.

The results indicate a positive association between securitisation market depth and bank lending. In the baseline specification, increases in Securitisation depth are associated with higher lending volumes, although the magnitude is modest. Additional analyses suggest that securitisation is more closely related to non-corporate lending and that the relationship is stronger in countries with structurally deeper securitisation markets.

The empirical strategy identifies conditional associations rather than causal effects. Although the specification controls for bank and year fixed effects as well as observable balance sheet characteristics, changes in securitisation activity may still be correlated with unobserved country-level developments that affect lending. The results should therefore be interpreted with this limitation in mind.

The remainder of the thesis proceeds as follows. Section 2 reviews the theoretical and empirical literature. Section 3 describes the data and main variables and provides descriptive evidence. Section 4 presents the conceptual framework on securitisation market depth and bank lending. Section 5 outlines the econometric framework and reports the empirical results, including robustness and heterogeneity analyses. Section 6 concludes and discusses policy implications.

2. Literature Review

The literature on securitisation and bank lending identifies several mechanisms through which securitisation may influence bank behaviour. Broadly, these mechanisms can be grouped into three categories: (i) risk transfer and secondary-market liquidity, (ii) regulatory capital and balance sheet constraints, and (iii) incentive distortions associated with originate-to-distribute models.

Theoretical Channels

A central theoretical mechanism concerns the extent to which securitisation renders otherwise illiquid bank assets marketable and redistributes risk through structured claims. Early work on loan sales highlights that bank loans are not naturally tradable because origination involves screening and monitoring, which creates informational asymmetries once loans are sold. G. B. Gorton and Pennacchi (1995) analyse loan sales in this setting and show how contractual design can mitigate moral hazard and sustain trade in bank-originated loans.

Related work studies the pricing and risk characteristics of structured credit. Duffie and Gârleanu (2001) develop a model of collateralised debt obligations (CDOs) in which tranching and default correlation determine valuation and risk allocation across investors. Their framework formalises how securitisation can redistribute risk toward investors with different risk-bearing capacities. Together, these contributions support the view that securitisation can transform the risk profile of underlying credit exposures and alter how risk is allocated across balance sheets.

A second strand emphasises regulatory capital and funding constraints. Pennacchi (1988) shows that loan sales can reduce the effective cost of bank capital, linking securitisation incentives to regulatory treatment. More broadly, banks face joint constraints related to capital regulation and liquidity provision. Kashyap et al. (2002) argue that deposit-taking and lending commitments create synergies that give banks a comparative advantage in liquidity provision. In this setting, securitisation may ease balance-sheet pressures by altering asset composition and funding structure, thereby affecting lending capacity.

Theoretical work also highlights potential distortions. Financial innovation may increase fragility if tail risks are neglected or shifted into securities perceived as safe. Gennaioli et al. (2012) develop a model in which neglected tail risks and financial innovation can generate systemic fragility. In addition, originate-to-distribute models may weaken

screening incentives if credit risk is transferred under asymmetric information. Finally, securitised banking structures that rely on short-term wholesale funding can be vulnerable to run-like dynamics; G. Gorton and Metrick (2012) analyse this mechanism in the context of repo markets during the 2007–2009 crisis.

Evidence from the United States

Empirical research, largely based on U.S. data, shows that securitisation affects both the level and allocation of credit. Loutskina and Strahan (2009) demonstrate that funding constraints matter less for loans that can be securitised, exploiting differences between jumbo and conforming mortgages to capture variation in secondary-market liquidity. Banks with greater access to securitisation are less sensitive to deposit supply shocks, consistent with securitisation providing an alternative funding channel. Loutskina (2011) further document that securitisation improves liquidity management and reduces the sensitivity of lending to funding shocks.

Another strand examines credit expansion and screening incentives during the pre-crisis mortgage boom. Mian and Sufi (2009) show that areas experiencing stronger mortgage credit growth between 2002 and 2005 subsequently faced higher default rates, particularly in regions with a high share of subprime borrowers. Their findings suggest that credit expansion was not primarily driven by improvements in fundamentals. Keys et al. (2010) exploit discontinuities in securitisation eligibility around FICO score thresholds and find that loans more likely to be securitised exhibited weaker ex post performance, consistent with reduced screening incentives.

Evidence from the crisis period further indicates that securitisation did not necessarily imply full risk transfer. Acharya et al. (2013) analyse asset-backed commercial paper conduits and document that sponsoring banks provided liquidity guarantees that effectively retained risk on their balance sheets. When conduits came under stress, losses largely reverted to sponsoring banks, suggesting that securitisation structures often involved limited effective risk transfer despite favourable regulatory treatment.

Evidence from Europe

Compared to the United States, empirical evidence on securitisation in Europe is more limited and institutionally distinct. Altunbas et al. (2009) show that European banks more active in securitisation are less sensitive to monetary policy tightening, consistent

with securitisation mitigating funding constraints and weakening the traditional bank lending channel.

At the bank level, Casu et al. (2013) document associations between securitisation activity, profitability, and risk exposure, indicating that securitisation is linked to broader changes in banks' balance-sheet structure and performance.

Post-crisis European securitisation markets have evolved under significant regulatory reform. Policy initiatives aimed at revitalising securitisation within the Capital Markets Union framework are discussed in European Commission (2015), while European Central Bank (2018) provide a detailed institutional overview of euro area securitisation markets. These accounts document subdued issuance volumes, changes in regulatory treatment, and a market structure that differs from the pre-crisis environment.

The existing literature provides extensive evidence on securitisation at the bank level, particularly in the United States. However, comparatively little work examines how differences in national securitisation market development relate to lending behaviour in the post-crisis euro area. By focusing on country-level securitisation market depth rather than bank-specific issuance, this thesis relates bank lending outcomes to broader institutional variation across euro-area member states. The following sections build on this framework to examine whether such variation is systematically associated with differences in lending levels and composition.

3. Data and Main Variables

3.1. Data Sources and Sample Construction

To examine whether securitisation activity is associated with bank lending in the euro area, I construct a bank-year panel dataset that combines bank-level balance sheet information from Bureau van Dijk's BankFocus database with country-level securitisation statistics from the European Central Bank (ECB). The sample covers the period 2009–2022.

Bank-level financial data are obtained from BankFocus, which provides harmonised annual financial statements for banks across European jurisdictions. The database includes detailed balance sheet and income statement information, including disaggregated loan categories, capital ratios, and funding structure variables widely used in the empirical banking literature (Altunbas et al., 2009; Casu et al., 2013). The unit of observation is the bank-year, identified using the unique Bureau van Dijk bank identifier.

BankFocus reports financial statements at different consolidation levels. Since this analysis focuses on lending decisions and balance sheet constraints at the level of banking groups, consolidated financial statements are prioritised. When multiple consolidation levels are available for a given bank–year, fully consolidated statements (C1) are selected first. If no C1 statement is available, partially consolidated statements (C2) are used. Other consolidation types are retained only when no higher-quality observation exists. This procedure balances data consistency with sample coverage and follows common practice in empirical banking research.

After applying this consolidation hierarchy, some bank–year observations remain duplicated. To ensure a single observation per bank–year, duplicates are resolved by retaining the observation with the highest data completeness, measured by the number of non-missing balance sheet and income statement variables. This yields the most comprehensive information available for each bank–year.

The sample is restricted to banks headquartered in euro-area countries over the period 2009–2022. The starting year excludes the most acute phase of the Global Financial Crisis and aligns with consistent securitisation data availability. To reduce noise from very small institutions, banks are required to have total assets exceeding 10 million. In addition, each bank must report at least three non-missing observations for the main lending variable.

3.1.1. Bank Lending Measures

The primary dependent variable is total bank lending. Total loans are constructed as the sum of mortgage loans, consumer loans, corporate loans, and other loans reported in BankFocus. These categories capture core on-balance-sheet lending activities and provide a comprehensive measure of credit provision.

In constructing total loans, missing loan components are treated as zero only if at least one other loan category is reported in the same bank–year. This approach assumes that unreported components reflect true zeros when other lending categories are disclosed. If all loan components are missing, total loans are coded as missing to avoid imputing lending activity where reporting is incomplete.

To account for differences in bank size and scale, total loans are transformed using the natural logarithm. The resulting variable, *log Total Loans*, serves as the dependent variable in the baseline regressions. Observations with zero total loans are excluded

from log specifications.

To examine heterogeneity across borrower types, I construct disaggregated lending measures. Corporate lending is defined as *log Corporate Loans*. Non-corporate lending is defined as total loans minus corporate loans, capturing lending to households and other non-corporate borrowers, and is subsequently transformed using the natural logarithm (*log Non-Corporate Loans*). These outcome variables allow me to assess whether securitisation is differentially associated with lending to firms versus households. Securitisation is typically more prevalent in standardised and homogeneous loan markets, most notably residential mortgages, where pooling and tranching are more feasible (Keys et al., 2010; Loutskina, 2011).

Because loan data availability varies across banks and over time, I construct an additional control variable measuring the number of loan components reported in a given bank–year. This variable ranges from zero to four and captures variation in reporting coverage that could otherwise mechanically affect measured lending volumes.²

3.1.2. Bank-Level Control Variables

To account for heterogeneity in bank characteristics that may influence lending behaviour, I include a standard set of bank-level control variables. Bank size is measured as *log Total Assets*. Larger banks typically have greater access to wholesale funding markets, more diversified income streams, and potentially different regulatory treatment, all of which may affect lending capacity and risk-taking.

Bank capitalisation is measured in the baseline specification using the equity-to-assets ratio. This leverage-based measure captures the share of assets financed by equity and serves as an indicator of balance sheet strength and loss-absorbing capacity. The equity-to-assets ratio is widely used in the empirical banking literature as a proxy for capital adequacy (Jiménez et al., 2012; Khwaja & Mian, 2008).

As a robustness check, I alternatively employ the Tier 1 capital-to-risk-weighted-assets ratio. The Tier 1 ratio reflects regulatory capital under the Basel framework and incorporates risk weights assigned to different asset classes. While conceptually closer to binding regulatory constraints, Tier 1 data are available only for a subset of banks, which reduces the effective sample size.

²Very small institutions are excluded to reduce noise from irregular reporters and entities with limited lending activity. Results are robust to alternative asset thresholds.

Funding structure is proxied by the deposit-to-assets ratio, capturing the extent to which a bank relies on stable retail funding relative to wholesale sources. Asset quality is measured using the non-performing loan (NPL) ratio, defined as non-performing loans divided by total loans. Higher NPL ratios may constrain new lending through capital impairment or heightened risk aversion. Profitability is captured by return on assets (ROA), defined as net income divided by total assets, reflecting internal capital generation and lending capacity.

All bank-level control variables enter the regressions with a one-year lag. This timing structure relates lending in year t to bank characteristics observed at the end of year $t - 1$, mitigating simultaneity concerns and reducing mechanical correlations between lending and contemporaneous balance sheet adjustments. To limit the influence of extreme observations, ratio variables are winsorised at the 1st and 99th percentiles.

3.1.3. Country-Level Securitisation Activity

Country-level securitisation data are obtained from the European Central Bank's Financial Vehicle Corporation (FVC) statistics.³ Rather than focusing on annual issuance flows, I use the outstanding stock of securitised assets. The stock measure captures the scale and structural importance of securitisation markets in which banks operate. Issuance volumes tend to be volatile and sensitive to short-term market conditions, regulatory timing, and deal-specific factors. By contrast, the outstanding stock reflects accumulated market depth and persistence of securitisation activity, making it more suitable for analysing medium-run lending behaviour.

For each country and year, securitisation stock is measured using end-of-year (fourth-quarter) observations. To ensure comparability across countries of different economic size, the stock of securitised assets is scaled by nominal GDP. Both variables are measured in million euros, and the resulting ratio represents securitisation depth relative to the size of the domestic economy. The empirical analysis uses the natural logarithm of Securitisation/GDP, lagged by one year. This lag structure reflects the idea that changes in the securitisation environment may influence bank lending decisions with a delay and mitigates concerns about contemporaneous feedback from lending to securitisation activity.

³Financial Vehicle Corporations (FVCs) are special-purpose entities used to hold securitised assets. The ECB collects harmonised balance-sheet data for FVCs across euro-area countries. See the ECB Statistical Data Warehouse: <https://data.ecb.europa.eu/data/datasets/FVC>.

Macroeconomic control variables are obtained from Eurostat. Nominal GDP is used to construct the securitisation-to-GDP ratio, while real GDP growth and the unemployment rate are included as country-level controls capturing aggregate demand conditions and macroeconomic risk. All macroeconomic variables enter the regressions with a one-year lag.

The FVC statistics include both placed and retained securitisations. The measure therefore captures the overall scale of securitisation activity, including transactions retained on bank balance sheets for funding or collateral purposes.⁴ Accordingly, Securitisation/GDP reflects the depth of securitisation markets and associated funding infrastructure rather than pure credit risk transfer.

Measuring securitisation activity at the country level reflects the institutional reality that securitisation markets operate largely through national legal, regulatory, and investor infrastructures. Differences in bankruptcy law, supervisory implementation, tax treatment, and the composition of domestic investor bases create country-specific securitisation environments to which banks within a jurisdiction are jointly exposed. Each bank is therefore assigned the Securitisation/GDP of its country of headquarters.

A further motivation for the country-level approach is that bank-level securitisation variables in BankFocus, such as *managed securitised assets* and *other exposure to securitisations*, exhibit extremely limited coverage in the sample (see Appendix Table A3). For the vast majority of bank–year observations, these variables are either unreported or missing. Relying on them would substantially reduce the effective sample size and introduce potential selection bias, as reporting banks may systematically differ from non-reporting institutions. Using harmonised ECB FVC data avoids these limitations and provides a consistent measure of the securitisation environment across countries and over time.

The empirical strategy exploits both cross-country differences and within-country time variation in Securitisation/GDP. While this approach does not isolate exogenous shocks to securitisation activity, it enables a systematic assessment of whether deeper securitisation markets are associated with higher subsequent bank lending, conditional on bank fixed effects, year fixed effects, and time-varying controls.

⁴In the European context, retained securitisations are frequently used as collateral in Eurosystem refinancing operations. As such, the measure captures funding-related securitisation activity even when credit risk is not fully transferred to external investors.

3.2. Summary Statistics

Table 1 reports summary statistics for the main variables used in the empirical analysis. The final sample consists of 8,292 bank-year observations for euro-area banks over the period 2009–2022. Due to differences in reporting coverage and the inclusion of lagged variables, the number of observations varies across individual variables.

The main dependent variable, $\log(\text{Total Loans})$, has a mean of 22.07 and a median of 22.51. These values correspond to total lending volumes of approximately 4–6 billion. Despite the logarithmic transformation, substantial cross-sectional heterogeneity remains.

Disaggregated lending measures reveal meaningful differences across loan types. Corporate lending exhibits a lower mean (19.94) and greater dispersion (standard deviation 3.05), reflecting the concentration of corporate credit among larger institutions. Non-corporate lending displays a higher mean (22.13) and somewhat narrower dispersion, suggesting a broader distribution across banks.

Bank size, measured by $\log(\text{Total Assets})$, has a mean of 22.94 and a median of 23.07, corresponding to balance sheet sizes of approximately 9–10 billion. Capitalisation also varies markedly. The equity-to-assets ratio has a mean of 12.2% and a median of 8.7%. Tier 1/RWA displays similar heterogeneity, with a median of 15.8% and upper-quartile values exceeding 20%, albeit with fewer observations.

Funding structures are also diverse. The median deposit-to-assets ratio is 55.5%, suggesting that customer deposits are the primary funding source for the typical bank. However, banks in the lower quartile rely on deposits for less than 30% of assets, indicating greater exposure to wholesale funding markets.

Asset quality, proxied by the non-performing loan (NPL) ratio, also varies substantially. The median bank reports an NPL ratio of 3.3%, while upper-quartile values exceed 6.1%. Profitability, measured by return on assets (ROA), has a mean of 0.6% and a median of 0.4%, consistent with the prolonged period of weak profitability in European banking following the Global Financial Crisis.

The key explanatory variable, *Securitisation/GDP*, exhibits substantial skewness across countries and over time. While the median stock-to-GDP ratio is 1.54, the mean is considerably higher at 3.44, reflecting the presence of countries with very deep securitisation markets. The interquartile range spans from 0.46 to 2.09, highlighting pronounced cross-country heterogeneity. To address skewness and facilitate interpretation,

the empirical analysis uses the natural logarithm of Securitisation/GDP. The transformed variable exhibits a more symmetric distribution while preserving economically meaningful variation.

Table 1: Summary Statistics

	Mean	Median	1st Qu.	3rd Qu.	SD	Obs.
<i>Lending Variables:</i>						
log(Total Loans)	22.067	22.514	20.597	23.741	2.655	8292
log(Corporate Loans)	19.939	20.465	17.553	22.185	3.053	4178
log(Non-Corporate Loans)	22.134	22.686	20.728	23.715	2.706	4171
<i>Bank Characteristics:</i>						
log(Total Assets)	22.944	23.074	21.539	24.268	2.174	7567
Equity-to-Assets Ratio	0.122	0.087	0.060	0.128	0.130	7565
Tier 1 / RWA Ratio	0.185	0.158	0.125	0.204	0.107	4368
Deposit-to-Assets Ratio	0.501	0.555	0.294	0.714	0.262	7158
NPL Ratio	0.055	0.033	0.020	0.061	0.069	1738
Return on Assets (ROA)	0.006	0.004	0.002	0.009	0.016	7054
<i>Securitisation and Macro Variables:</i>						
Securitisation/GDP	3.436	1.539	0.455	2.091	8.731	7010
log(Securitisation/GDP)	0.209	0.421	-0.787	0.738	1.357	6307
GDP Growth (%)	0.988	1.317	0.216	2.259	3.525	6345
Unemployment Rate (%)	9.190	8.600	5.900	10.300	4.661	6345

Macroeconomic conditions also vary widely over the sample period. GDP growth averages approximately 1% but displays substantial dispersion. The unemployment rate has a median of 8.6%, with upper-quartile values exceeding 10%, reflecting persistent heterogeneity in labour market conditions across euro-area economies.

3.3. Descriptive Evidence on Securitisation and Lending in the Eurozone

To establish stylised facts about the post-crisis evolution of European securitisation markets and to highlight the cross-country heterogeneity that motivates the panel-data strategy, this section documents the development of Securitisation/GDP and aggregate bank lending in the euro area. The descriptive evidence highlights three patterns: (i) a pronounced and persistent decline in aggregate securitisation depth following the Global Financial Crisis, (ii) substantial and persistent cross-country differences

in securitisation intensity, and (iii) lending dynamics that do not mechanically mirror aggregate securitisation trends.

Figure 1 shows aggregate securitisation stock relative to GDP for the euro area over the period 2009–2022. I construct the aggregate series by summing country-level outstanding securitised assets from the ECB’s FVC statistics and scaling by aggregate euro-area GDP. The figure reveals a pronounced and sustained contraction in securitisation activity in the early 2010s. The Securitisation/GDP ratio declines from above 23% in 2009 to approximately 13–14% by the mid-2010s. This adjustment coincides with the aftermath of the Global Financial Crisis and the European sovereign debt crisis—periods characterised by elevated risk aversion, balance-sheet deleveraging, regulatory tightening, and fragmentation in euro-area financial markets. The pattern is consistent with the broader disruption of securitised funding markets documented by G. Gorton and Metrick (2012).

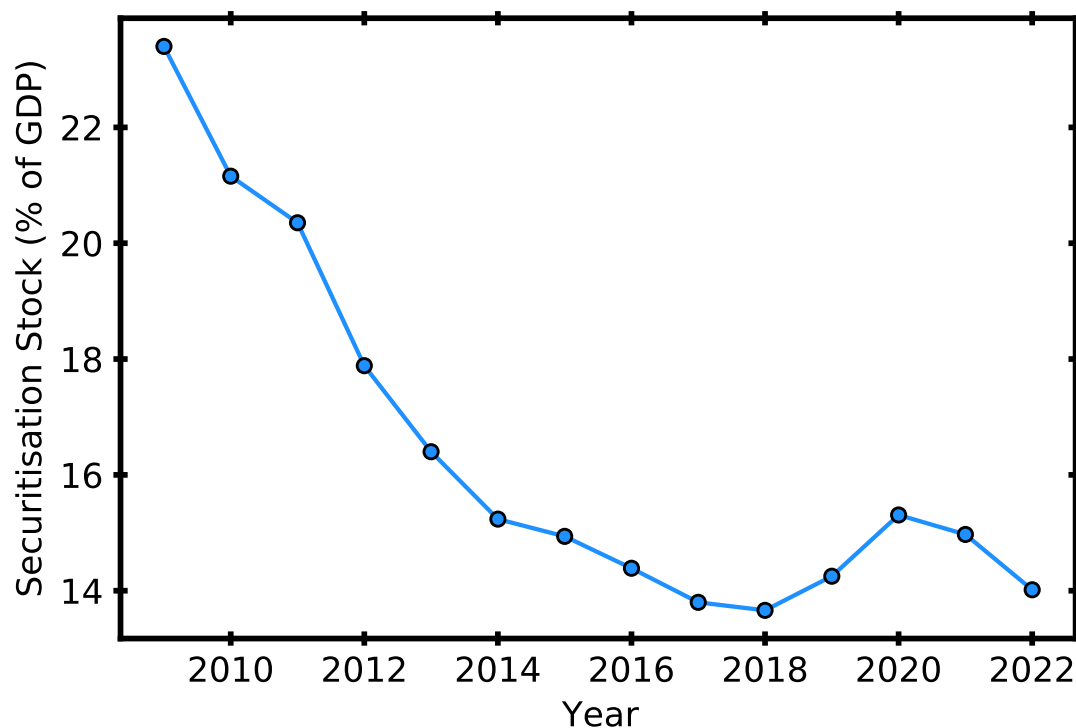


Figure 1: Securitisation/GDP in the Eurozone (2009–2022)

In the European context, post-crisis reforms likely altered issuance incentives and reshaped the institutional structure of securitisation markets. These include stricter capital and liquidity requirements under Basel III⁵ as well as mandatory risk-retention rules and

⁵The Basel III framework strengthened minimum capital requirements, introduced a leverage ratio, and

enhanced disclosure standards under the EU Securitisation Regulation⁶. Together with macro-financial adjustments, these changes appear to have contributed to a structurally lower level of securitisation relative to GDP.

From around 2017 onward, securitisation activity stabilises and exhibits a modest recovery, with a temporary uptick around 2020. This period coincides with exceptionally accommodative monetary policy and renewed policy emphasis on securitisation within the Capital Markets Union framework. Nevertheless, by the end of the sample period, Securitisation/GDP remains substantially below its immediate post-crisis level, suggesting that securitisation has not regained its pre-crisis systemic importance within the euro-area financial architecture.

Two implications follow. First, securitisation in Europe appears to have undergone a persistent post-crisis contraction rather than a cyclical downturn followed by a full recovery. Second, despite the overall downward shift, meaningful time-series variation remains, which is essential for identification in panel regressions exploiting within-country changes over time.

While aggregate developments are informative, they conceal substantial cross-country heterogeneity. Figure 2 presents Securitisation/GDP for selected euro-area countries (Spain, the Netherlands, Italy, France, and Germany) in two benchmark years, 2010 and 2019. The figure highlights striking cross-country differences. In 2010, Securitisation/GDP exceeded 60% in the Netherlands and 40% in Spain, whereas Germany exhibited persistently low levels, consistent with its traditionally bank-based financial system and limited reliance on securitisation markets.

Between 2010 and 2019, securitisation activity declined in all countries shown, although the magnitude of the decline varies. Spain and the Netherlands experienced particularly sharp contractions, while France and Italy show more moderate declines. Germany remains persistently low throughout the period.

These patterns point to two well-documented features of European financial systems. First, securitisation activity was concentrated in a subset of countries prior to the crisis. Second, post-crisis retrenchment was widespread but uneven. Cross-country differences in financial system structure and market development are documented in the broader

established liquidity standards such as the Liquidity Coverage Ratio (LCR) and the Net Stable Funding Ratio (NSFR); see the Basel Committee on Banking Supervision (2011, 2013).

⁶Regulation (EU) 2017/2402 established a general framework for securitisation and introduced the Simple, Transparent and Standardised (STS) label, along with harmonised due diligence and risk-retention requirements.

literature (Rajan & Zingales, 1998). Such institutional differences are likely to shape the depth and functioning of securitisation markets across jurisdictions.

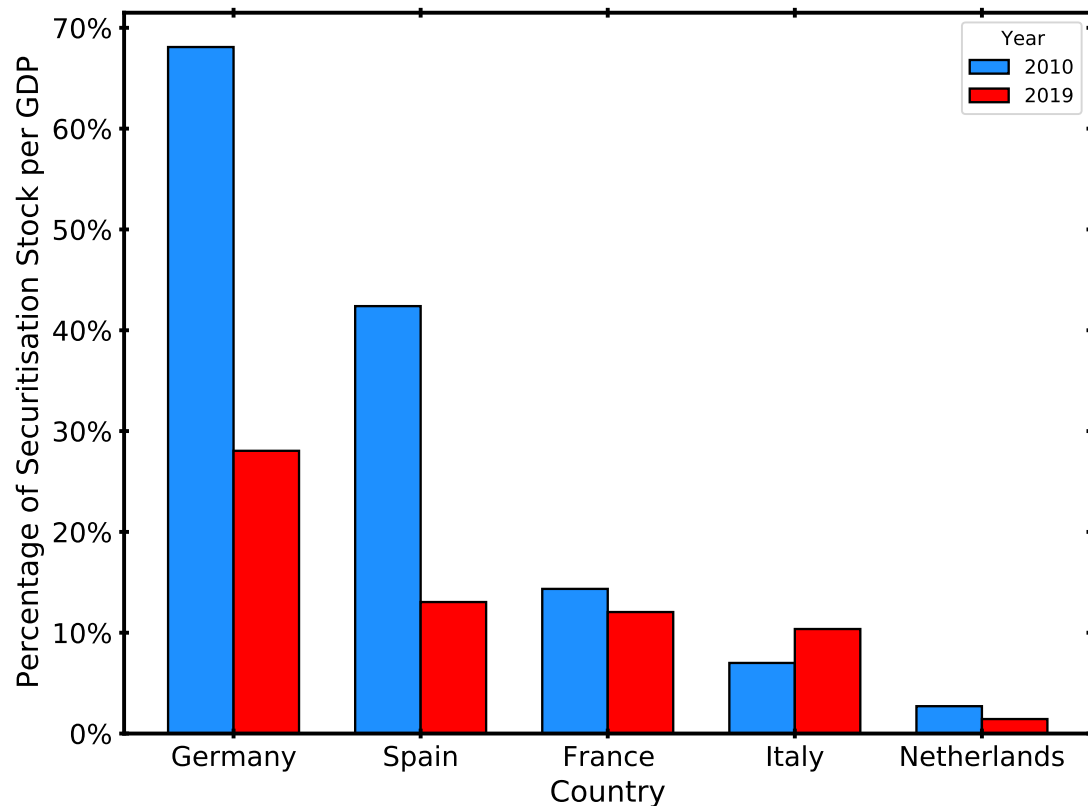


Figure 2: Securitisation/GDP in Selected Countries (2010 and 2019)

In contrast to the sharp contraction in securitisation activity, aggregate bank lending follows a more gradual and uneven trajectory. Figure 3 shows the evolution of aggregate bank lending in the euro area, measured as total loans summed across banks and expressed in trillion euros.⁷ Lending rises in the early part of the sample, flattens somewhat in the mid-2010s, and then resumes stronger growth toward the end of the period.

Notably, aggregate lending expands despite Securitisation/GDP remaining at historically subdued levels. The divergence between securitisation and aggregate lending suggests that securitisation is neither the sole nor the dominant determinant of euro-area lending dynamics. Banks appear to have adjusted to the post-crisis environment through alternative funding sources, balance sheet repair, capital accumulation, and monetary policy support.

The absence of a simple aggregate correlation reinforces the importance of a bank-level

⁷Total loans are converted into trillion euros by dividing by 10^{12} .

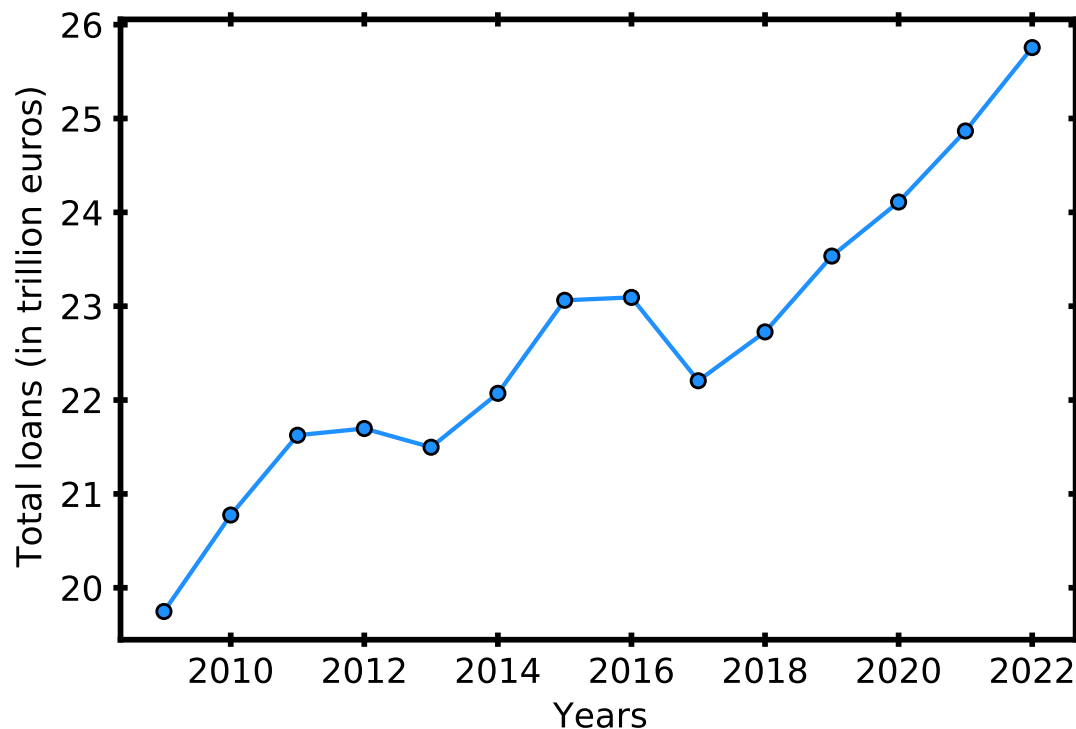


Figure 3: Aggregate Bank Lending in the Eurozone

panel approach that accounts for heterogeneity across institutions and countries. By exploiting within-bank variation over time and cross-country differences in Securitisation/GDP, the empirical strategy moves beyond aggregate correlations and enables a more granular assessment of the association between securitisation market depth and lending behaviour.

4. Securitisation Market Depth and Bank Lending

Banks extend credit subject to capital regulation, funding constraints, and liquidity management considerations. The extent to which securitisation markets are developed may influence how binding these constraints are and thereby affect lending behaviour.

A first mechanism operates through liquidity and funding conditions. Banks jointly manage lending and liquidity provision, as deposit-taking and credit commitments create interconnected balance-sheet exposures (Kashyap et al., 2002). When funding conditions tighten, lending may contract. However, the ability to securitise loans increases the liquidity of bank assets and allows institutions to convert otherwise illiquid exposures

into tradable claims. Empirical evidence from the United States shows that loans with greater secondary-market liquidity are significantly less sensitive to banks' funding shocks (Loutskina & Strahan, 2009). This suggests that deeper securitisation markets may weaken the link between funding constraints and credit supply.

A second mechanism relates to regulatory capital. Loan sales and securitisation can alter the effective cost of capital by transferring credit risk or reducing risk-weighted assets (Pennacchi, 1988). In practice, securitisation structures have at times provided regulatory capital relief even when risk transfer was incomplete (Acharya et al., 2013). To the extent that securitisation depth reflects the institutional infrastructure that enables such balance-sheet management, banks operating in countries with more developed securitisation markets may face looser effective capital constraints and thus greater lending capacity.

A third channel concerns loan composition. Not all loans are equally suitable for securitisation. Standardised and homogeneous loans are more easily pooled and sold than relationship-based credit. Evidence indicates that differences in loan liquidity translate into differences in credit supply sensitivity (Loutskina & Strahan, 2009). Deeper securitisation markets may therefore be associated not only with higher aggregate lending but also with shifts in the composition of credit toward more securitisable categories.

Collectively, these mechanisms imply that securitisation market depth can affect lending through funding liquidity, regulatory capital, and asset composition channels. However, the net effect is theoretically ambiguous. While securitisation may relax balance-sheet constraints and expand credit supply, it may also alter screening incentives or concentrate activity in specific loan segments.

5. Regression Analysis

To examine whether variation in securitisation activity is systematically associated with bank lending in the euro area, the empirical analysis builds on established theoretical mechanisms linking secondary loan markets to bank balance-sheet constraints. Prior research shows that access to securitisation and loan sales can reduce the sensitivity of lending to funding shocks and regulatory capital constraints by improving asset liquidity and facilitating risk transfer (Kashyap et al., 2002; Loutskina & Strahan, 2009). In the post-crisis European environment, however, securitisation markets were subject to substantial regulatory reform, increased risk retention requirements, and persistent

financial fragmentation. Whether securitisation continues to support lending under these altered institutional conditions is therefore an empirical question.

In the baseline specification, I hypothesise that deeper securitisation markets are associated with higher subsequent lending by banks operating in the respective jurisdiction. If securitisation enhances funding flexibility or relaxes effective capital constraints, increases in Securitisation/GDP should coincide with expansions in lending, conditional on bank characteristics and common macro-financial conditions.

The empirical strategy exploits within-bank variation over time in exposure to changes in the domestic securitisation environment. This approach follows standard panel-data methods in the credit supply literature, where bank-level outcomes are related to institutional or regulatory variation while controlling for borrower- and bank-specific heterogeneity (Jiménez et al., 2012; Khwaja & Mian, 2008).

5.1. Econometric Model

The main explanatory variable is the logarithm of Securitisation/GDP. The central question is whether changes in this measure are associated with subsequent changes in bank lending. Because securitisation markets operate at the national level while lending decisions are observed at the bank level, the empirical framework links country–year variation to bank–year outcomes.

The specification controls for persistent differences across banks and common shocks affecting all institutions. Banks differ in size, business models, capital structures, and risk profiles, all of which may influence lending behaviour. At the same time, euro-area-wide developments, including monetary policy interventions, sovereign stress episodes, and regulatory reforms, affect all banks simultaneously. The baseline model therefore includes bank fixed effects and year fixed effects. The estimating equation is:

$$\log(\text{Total Loans}_{i,c,t}) = \alpha + \beta \log(\text{Securitisation/GDP}_{c,t-1}) + \mathbf{X}'_{i,t-1} \gamma + \alpha_i + \delta_t + \varepsilon_{i,c,t}, \quad (1)$$

where $\log(\text{Total Loans}_{i,c,t})$ denotes the natural logarithm of total loans of bank i in country c at time t . The key explanatory variable is the lagged logarithm of Securitisation/GDP at the country level. The coefficient of interest, β , measures the elasticity of bank lending with respect to securitisation market depth.

The vector $\mathbf{X}_{i,t-1}$ includes lagged bank-level control variables: bank size (log total assets), capitalisation (equity-to-assets ratio in the baseline and alternatively the Tier 1 capital ratio), funding structure (deposit-to-assets ratio), asset quality (non-performing loan ratio), and profitability (return on assets). Lagging these variables relates lending in year t to bank characteristics observed at the end of year $t - 1$, thereby mitigating concerns of simultaneity and mechanical balance-sheet correlations.

Bank fixed effects (α_i) absorb all time-invariant bank characteristics, including differences in long-run business models, ownership structures, and risk appetite. Year fixed effects (δ_t) capture shocks common to all banks in a given year, such as euro-area monetary policy changes, macroeconomic conditions, and regulatory shifts.

Identification arises from within-bank changes over time in exposure to country-level securitisation dynamics, after accounting for observable bank characteristics and common time effects.⁸ Standard errors are clustered at the country level to allow for arbitrary correlation of residuals among banks operating within the same jurisdiction.

5.1.1. Baseline Results

Table 2 reports the results of equation (1) using bank–year panel data. Column (5) presents the fully specified model, including bank and year fixed effects as well as the complete set of lagged bank-level controls.

The coefficient of $\log(\text{Securitisation}/\text{GDP})$ is 0.109 and statistically significant at the 5% level ($p = 0.012$). This estimate can be interpreted as an elasticity and suggests that a 1% increase in securitisation market depth is associated with an approximately 0.109% increase in bank lending.

In economic terms, this magnitude is meaningful given the subdued credit growth environment of the post-crisis euro area. For instance, a 50% increase in securitisation intensity would be associated with roughly a 5.5% increase in lending, holding other factors constant. While the specification does not identify a causal effect, the positive and statistically significant coefficient indicates a systematic conditional association between deeper securitisation markets and higher lending volumes.

The positive association is consistent with theoretical models showing how securitisation

⁸Although the specification exploits within-bank variation over time, it does not isolate plausibly exogenous variation in $\text{Securitisation}/\text{GDP}$. The estimated coefficients should therefore be interpreted as conditional associations rather than causal effects.

transforms illiquid loans into tradable claims, thereby affecting balance-sheet liquidity and risk allocation in the presence of informational frictions (e.g., G. B. Gorton and Pennacchi (1995)). Models of structured credit further emphasise how tranching redistributes correlated default risk across investors (Duffie and Gârleanu (2001)), mechanisms that may enhance funding flexibility. Empirically, U.S. evidence shows that access to securitisation markets is associated with reduced sensitivity of lending to funding constraints (Loutskina and Strahan (2009)). Although institutional differences imply that European markets may operate through distinct regulatory and funding channels, the direction of the estimated association is consistent with these mechanisms.

Turning to bank-level controls, the coefficient of $\log(\text{Total Assets})$ is 0.764 and highly statistically significant at the 0.1% level ($p < 0.001$). Interpreted as an elasticity, this implies that a 10% increase in total assets is associated with approximately a 7.6% increase in lending. The coefficient below unity indicates that marginal balance-sheet expansion is not allocated proportionally to loans, potentially reflecting diversification into other asset classes or the accumulation of liquidity buffers.

The equity-to-assets ratio enters with a coefficient of -4.607 and is statistically significant at the 1% level. Given that capital ratios vary within a relatively narrow range, a more economically relevant interpretation considers a 5 percentage point increase in the equity-to-assets ratio. This change is associated with an approximate 21% decline in lending.⁹

This negative association likely reflects balance-sheet adjustment dynamics rather than a mechanical contraction effect. In the post-crisis regulatory environment, increases in capital ratios were often achieved through deleveraging, asset-side adjustments, or shifts toward lower-risk asset compositions. The estimate is therefore consistent with periods of balance-sheet repair during which stronger capitalisation coincided with weaker loan growth. Moreover, evidence that securitisation may involve limited risk transfer and regulatory capital motives (Acharya et al. (2013)) underscores the need for caution when interpreting balance-sheet relief mechanisms as automatically expanding lending.

The coefficients on Deposit/Assets (0.261), NPL Ratio (0.284), and ROA (4.255) are not statistically significant in the fully specified model. The dummy variable *Loan components present* is positive and significant (0.134), indicating that banks reporting

⁹In a log-level specification, the exact percentage effect of a change Δx in a level variable is given by $e^{\beta \Delta x} - 1$. Using $\beta = -4.607$ and $\Delta x = 0.05$, this yields $e^{-0.230} - 1 \approx -0.206$, or about a 21% decrease in lending.

Table 2: Baseline regression results

	(1)	(2)	(3)	(4)	(5)
log(Securitisation/GDP)	0.231* (0.023)	0.056 (0.352)	0.074 (0.254)	0.107* (0.022)	0.109* (0.012)
log(Total Assets)		0.993*** (< 0.001)	0.927*** (< 0.001)	0.776*** (< 0.001)	0.764*** (< 0.001)
Equity/Assets			0.106 (0.932)	-4.592** (0.002)	-4.607** (0.002)
Deposit/Assets			0.757** (0.008)	0.321 (0.354)	0.261 (0.423)
NPL Ratio				0.210 (0.674)	0.284 (0.552)
ROA				4.399 (0.206)	4.255 (0.187)
Loan components present					0.134** (0.002)
Num. Obs.	6213	6213	5881	1349	1349
R^2	0.954	0.968	0.971	0.982	0.983
Within R^2	0.008	0.325	0.312	0.358	0.375
Std. Errors	by: country	by: country	by: country	by: country	by: country
Bank FE	X	X	X	X	X
Year FE	X	X	X	X	X

Notes: The dependent variable is the logarithm of total bank loans. The main explanatory variable is the lagged ratio of securitisation stock to GDP at the country–year level. All regressions include bank fixed effects and year fixed effects. All control variables are lagged by one period. Standard errors are clustered at the country level, the level at which the main explanatory variable varies. Numbers in parentheses report p -values. Statistical significance is indicated by *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

more detailed loan breakdowns have, on average, higher measured lending. This variable is intended to account for reporting heterogeneity rather than to capture a structural lending mechanism.

Columns (1)–(4) show how the estimated securitisation coefficient evolves as controls are introduced. In Column (1), which includes only fixed effects and securitisation, the coefficient on log(Securitisation/GDP) is 0.231 and statistically significant at the 5% level. Once log(Total Assets) is added in Column (2), the securitisation coefficient declines to 0.056 and becomes statistically insignificant. This pattern suggests that part of the unconditional association reflects differences in the size distribution of banks across countries and over time, rather than a stable conditional relationship between securitisation intensity and lending.

The strong role of bank characteristics in explaining lending outcomes is consistent with evidence that lending responses to financial conditions vary systematically with bank balance-sheet structure and size (Kashyap and Stein (2000)). In Column (3), the inclusion of funding structure variables leaves the securitisation coefficient positive but statistically insignificant (0.074). Although the Deposit/Assets coefficient is positive and significant in Column (3), it should be interpreted as a semi-elasticity: the estimated 0.757 implies that a 10 percentage point increase in Deposit/Assets is associated with approximately a 7.6% increase in lending.

In Column (4), controlling additionally for asset quality (NPL Ratio) and profitability (ROA), the securitisation coefficient increases to 0.107 and regains statistical significance at the 5% level, despite the reduction in sample size. This pattern suggests that omission of risk and performance measures may affect the estimated association between securitisation intensity and lending.

Collectively, the results suggest that although part of the unconditional association between securitisation and lending reflects bank size and balance-sheet structure, a positive conditional relationship persists once these factors are accounted for. The findings are consistent with the view that securitisation market depth remains relevant for lending outcomes in Europe in the post-crisis environment, while warranting cautious interpretation given regulatory change and the possibility of limited risk transfer (Loutskina and Strahan (2009); Acharya et al. (2013)).¹⁰

5.1.2. Robustness Checks: Loan Coverage and Reporting Consistency

A potential concern in the bank-level data is that variation in total lending may partly reflect differences in reporting practices rather than genuine changes in credit supply. Although the baseline specification controls for reporting intensity by including the number of reported loan components, I further address this issue by restricting the sample to bank–year observations with at least two reported loan categories.

The results are reported in Table A7 in the Appendix. The main findings remain unchanged. The coefficient of $\log(\text{Securitisation}/\text{GDP})$ remains positive and statistically significant in the fully controlled specification, and its magnitude is comparable to and

¹⁰As an alternative measure of capitalisation, I replace the equity-to-assets ratio with the Tier 1 capital-to-risk-weighted-assets ratio (see Appendix Table A6). The Tier 1/RWA ratio is statistically insignificant and economically small across specifications, suggesting that lending dynamics in this sample are more closely related to simple leverage than to risk-weighted regulatory capital measures.

slightly larger than that obtained in the baseline sample.

Similarly, the negative and statistically significant association between capitalisation and lending persists, and the elasticity of lending with respect to bank size remains stable across specifications. Overall, the pattern closely mirrors the baseline results.

These findings indicate that the positive association between securitisation activity and lending is not driven by sparse or inconsistent loan reporting. Restricting attention to bank–year observations with more comprehensive loan coverage does not weaken the results and strengthens confidence that the baseline estimates reflect underlying lending dynamics rather than reporting artefacts.

5.1.3. Robustness by Loan Type: Corporate versus Non-Corporate Lending

To examine whether the relationship between securitisation activity and bank lending differs across loan categories, I distinguish between corporate and non-corporate lending. Securitisation markets are typically more developed for retail and mortgage-type loans than for corporate credit. Moreover, information asymmetries, collateral structures, and monitoring requirements differ substantially across these loan types, implying that securitisation may affect them through distinct channels (Boot & Thakor, 2000; Holmstrom & Tirole, 1997).

The baseline specification is re-estimated separately for corporate and non-corporate lending. The estimating equation is identical to Equation (1), except that the dependent variable is replaced by loan type $k \in \{\text{Corporate}, \text{NonCorporate}\}$:

$$\log(\text{Loans}_{i,c,t}^k) = \alpha + \beta \log(\text{Securitisation/GDP}_{c,t-1}) + \mathbf{X}'_{i,t-1}\gamma + \alpha_i + \delta_t + \varepsilon_{i,c,t}. \quad (2)$$

Table 3 reports the results. Column (1) uses the logarithm of corporate loans as the dependent variable, while Column (2) uses the logarithm of non-corporate loans. All specifications include bank and year fixed effects, and standard errors are clustered at the country level.

In Column (1), the coefficient on $\log(\text{Securitisation/GDP})$ is -0.361 and statistically significant at the 10% level ($p = 0.064$). Interpreted as an elasticity, this estimate implies that a 10% increase in securitisation activity is associated with approximately a 3.6%

decrease in corporate lending. While the statistical significance is modest and warrants caution, the negative coefficient suggests that greater securitisation intensity may coincide with a relative contraction in corporate lending at the bank level.

One possible interpretation is that securitisation may induce a reallocation of bank balance sheets toward loan categories that are more standardised and easier to securitise, such as mortgages and consumer credit, and away from information-intensive corporate lending. Corporate loans typically rely heavily on relationship-specific screening and monitoring, making them less suitable for standardised securitisation structures. This interpretation is consistent with theoretical arguments emphasising banks' comparative advantage in relationship lending relative to market-based finance (Boot & Thakor, 2000; Holmstrom & Tirole, 1997).

Column (2) reports results for non-corporate lending, which primarily includes household-related loans such as mortgages and consumer credit. The coefficient on $\log(\text{Securitisation}/\text{GDP})$ is positive (0.090) but not statistically significant ($p = 0.508$). Although the lack of statistical precision precludes firm conclusions, the positive sign contrasts with the negative coefficient estimated for corporate lending.

In all, the results suggest that securitisation activity may be associated with shifts in the composition of lending rather than a uniform expansion across all loan types. However, given the reduced sample size and statistical uncertainty, these results should be interpreted as suggestive rather than conclusive evidence of compositional effects.

Across both specifications, $\log(\text{Total Assets})$ is positively and statistically significantly associated with lending, with a larger elasticity for non-corporate loans. The coefficient on $\text{Equity}/\text{Assets}$ is negative and statistically significant in both regressions, indicating that higher capital ratios are associated with lower loan volumes across loan categories. This pattern is consistent with balance-sheet adjustment mechanisms and evidence that capital structure dynamics often reflect asset-side contractions rather than equity issuance (Gropp & Heider, 2009).

These estimates indicate that securitisation does not uniformly expand all forms of bank lending. Instead, it may influence the composition of credit supply, with heterogeneous associations across loan types. This compositional interpretation is broadly consistent with U.S. evidence linking securitisation activity to retail mortgage lending dynamics (Loutskina & Strahan, 2009), while highlighting potentially distinct patterns within the European banking context.

Table 3: Robustness: Loan Type Heterogeneity

	Log Corporate Loans	Log Non-Corporate Loans
log(Securitisation/GDP)	-0.361. (0.064)	0.090 (0.508)
log(Total Assets)	0.560* (0.040)	0.858*** (< 0.001)
Equity/Assets	-4.173** (0.002)	-6.645*** (< 0.001)
Deposit/Assets	0.137 (0.802)	0.813 (0.122)
NPL Ratio	-0.396 (0.638)	0.826 (0.218)
ROA	5.258 (0.427)	5.480 (0.270)
Loan components present	0.138 (0.276)	0.288 (0.073)
Num. Obs.	1021	1027
R^2	0.974	0.968
Within R^2	0.091	0.344
Std. Errors	Clustered by country	
Bank FE	X	X
Year FE	X	X

Notes: The dependent variable is the logarithm of corporate loans in column (1) and the logarithm of non-corporate loans in column (2). The key explanatory variable is the lagged logarithm of Securitisation/GDP at the country–year level. All regressions include bank fixed effects and year fixed effects. Control variables are lagged by one year. Standard errors are clustered at the country level. Numbers in parentheses report p -values. Statistical significance is indicated by *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, . $p < 0.1$.

5.2. Heterogeneity Across Banks

The baseline results indicate a positive average association between securitisation activity and bank lending. However, theory suggests that financial innovations rarely affect all institutions in the same way. Banks differ in their capital strength, risk exposure, and overall balance-sheet structure. It is therefore plausible that securitisation matters more for some banks than for others.

To explore this possibility, I examine whether the relationship between Securitisation/GDP and lending varies across banks. In particular, I focus on two dimensions that are central in the banking literature: capital strength and credit risk. The aim is to assess whether securitisation is more strongly associated with lending for banks that appear more constrained or more vulnerable.

A well-established insight in the literature is that banks' lending responses depend on their balance-sheet position. Gertler and Kiyotaki (2010) argue that intermediaries with weaker capital positions are more sensitive to changes in liquidity conditions, as limited net worth constrains their ability to expand lending. Similarly, Khwaja and Mian (2008) and Jiménez et al. (2012) show that bank balance-sheet strength shapes lending responses to shocks. If securitisation relaxes funding or capital constraints, one might expect its effects to be stronger for banks that are relatively weaker. At the same time, post-crisis regulation and market discipline may limit the extent to which such banks can expand lending through securitisation-related channels.

To test these ideas, I extend the baseline specification by interacting Securitisation/GDP with bank-level dummy variables capturing weak capitalisation and high credit risk.

Bank capital strength is measured using the equity-to-assets ratio. For each year, I define a dummy variable equal to 1 for banks with an equity-to-assets ratio below the cross-sectional median, and 0 otherwise. Using a within-year median split ensures that the classification reflects relative capital strength in a given year rather than time trends driven by regulatory reforms or macroeconomic developments.¹¹

Bank risk exposure is proxied by the non-performing loan (NPL) ratio. A high-risk dummy variable is defined as 1 for banks with an NPL ratio above the within-year median, and 0 otherwise. This ensures that the measure captures cross-sectional differences in credit risk rather than aggregate movements in loan quality.

¹¹ Using a median-based dummy provides a transparent classification. Results are qualitatively similar when defining weakly capitalised banks as those in the bottom tercile (see Appendix Table A8).

The estimating equation is:

$$\begin{aligned} \log(\text{Total Loans}_{i,c,t}) = & \alpha + \beta_1 \log(\text{Securitisation/GDP}_{c,t-1}) \\ & + \beta_2 \left[\log(\text{Securitisation/GDP}_{c,t-1}) \times D_{i,t-1} \right] \\ & + \beta_3 H_{i,t-1} + \mathbf{X}'_{i,t-1} \gamma + \alpha_i + \delta_t + \varepsilon_{i,c,t}, \end{aligned} \quad (3)$$

where $H_{i,t-1}$ denotes either the weak-capital dummy or the high-risk dummy. The coefficient β_1 captures the association between securitisation and lending for the reference group, while β_2 measures whether this association differs for the respective group of banks.

5.2.1. Heterogeneity by Bank Capital Strength

Column (1) of Table 4 reports the results when interacting $\log(\text{Securitisation/GDP})$ with the weak-capital dummy. The coefficient on $\log(\text{Securitisation/GDP})$ is 0.119 ($p = 0.013$), representing the elasticity for better-capitalised banks. A 1% increase in $\text{Securitisation/GDP}$ is associated with a 0.119% increase in lending for this group.

The interaction term is economically small (0.003) and statistically insignificant ($p = 0.842$). This suggests that the association between securitisation market depth and lending does not differ meaningfully between weaker and stronger banks in terms of capital. Although the weak-capital dummy itself is negative (-0.111 , $p = 0.057$), indicating lower average lending among relatively weaker banks, there is no evidence that securitisation disproportionately coincides with stronger lending by undercapitalised institutions.

Overall, these results do not support the view that securitisation primarily operates as a mechanism through which capital-constrained banks expand credit.

5.2.2. Heterogeneity by Bank Risk

Column (2) of Table 4 examines heterogeneity by bank risk. The coefficient on $\log(\text{Securitisation/GDP})$ remains positive and statistically significant (0.097, $p = 0.018$), capturing the association for relatively low-risk banks.

The interaction between $\log(\text{Securitisation/GDP})$ and the high-risk dummy is positive

Table 4: Heterogeneity Analysis

	(1)	(2)
log(Securitisation/GDP)	0.119* (0.013)	0.097* (0.018)
Weak Capital (Dummy)	-0.111. (0.057)	
log(Total Assets)	0.795*** (< 0.001)	0.764*** (< 0.001)
Equity/Assets	-4.857** (0.002)	-4.614** (0.002)
Deposit/Assets	0.254 (0.410)	0.271 (0.386)
NPL Ratio	0.265 (0.572)	0.309 (0.482)
ROA	4.453 (0.154)	4.169 (0.175)
Loan components present	0.126** (0.004)	0.134** (0.002)
log(Securitisation/GDP) $\times$ Weak Capital	0.003 (0.842)	
High Risk (Dummy)		0.031 (0.737)
log(Securitisation/GDP) $\times$ High Risk		0.020 (0.649)
Num. Obs.	1349	1349
R^2	0.983	0.983
Within R^2	0.381	0.375
Std. Errors	by country	by country
Bank FE	X	X
Year FE	X	X

Notes: Weak-capital banks are defined as banks with lagged equity-to-assets ratios below the within-year median (dummy equals one). High-risk banks are defined as banks with lagged NPL ratios above the within-year median (dummy equals one). All regressions include bank fixed effects and year fixed effects. Control variables are lagged by one year. Standard errors are clustered at the country level. Numbers in parentheses report p -values. Statistical significance is indicated by *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, . $p < 0.1$.

(0.020) but statistically insignificant ($p = 0.649$). The high-risk dummy itself is also insignificant. Taken together, these results indicate that the association between securitisation market depth and lending does not differ meaningfully between higher-

and lower-risk banks.

From a financial stability perspective, this is noteworthy. While some studies suggest that securitisation can weaken screening incentives or alter risk-taking behaviour (Keys et al. (2010)), the evidence here does not indicate that riskier banks expand lending more strongly when securitisation activity increases. In the euro-area setting examined in this thesis, securitisation appears to be associated with lending in a broadly similar manner across institutions with different levels of observed credit risk.¹²

5.3. Country-Level Evidence

The baseline analysis relies on within-bank variation over time. However, securitisation markets are shaped primarily at the national level. Legal frameworks, supervisory practices, investor participation, and market infrastructure differ across countries and may influence how effectively banks can originate, structure, and distribute securitised assets (Levine, 2005; Rajan & Zingales, 1998).

Even if individual banks differ in their balance sheets, the broader domestic securitisation environment may affect credit supply through institutional and financial development channels. I therefore complement the bank-level evidence with a country-level analysis that relates changes in Securitisation/GDP to aggregate lending.

This extension serves two purposes. First, it provides a macro-level perspective that complements the micro-level regressions. Second, it allows the securitisation–lending relationship to vary across countries with structurally different levels of securitisation market development.

To construct the country–year panel, I aggregate total loans across all banks in each country and year:

$$\text{Total Loans}_{c,t} = \sum_{i \in c} \text{Total Loans}_{i,c,t}. \quad (4)$$

The aggregation includes all banks satisfying the baseline sample restrictions described in Chapter 3. In addition, I record the number of active banks in each country–year to track changes in market structure. Summary statistics for the country–year panel are reported in Appendix Table A4.

¹²Results are robust to alternative dummy definitions based on terciles of the capital and NPL distributions (see Appendix Table A8).

Aggregate lending is transformed using the natural logarithm, consistent with the bank-level specification. This transformation also reduces skewness and limits the influence of very large banking systems.

The lending aggregates are then merged with country-level securitisation and macroeconomic data. As before, securitisation activity is measured as the stock of securitised assets relative to GDP. Macroeconomic controls include lagged real GDP growth and lagged unemployment, which proxy for demand conditions and macroeconomic risk (Bernanke & Gertler, 1995). All explanatory variables enter with a one-year lag. The final panel covers euro-area countries over 2009–2022.

The baseline country-level specification is:

$$\log(\text{Total Loans}_{c,t}) = \alpha + \beta \log(\text{Securitisation/GDP}_{c,t-1}) + \gamma \mathbf{X}_{c,t-1} + \alpha_c + \lambda_t + \varepsilon_{c,t}, \quad (5)$$

where α_c and λ_t denote country and year fixed effects. Identification relies on within-country time variation. The coefficient β captures whether increases in Securitisation/GDP are associated with changes in aggregate lending, conditional on macroeconomic conditions and common shocks.

To capture persistent cross-country differences in securitisation development, I construct a structural securitisation-depth measure:

$$\overline{\text{SecDepth}}_c = \frac{1}{T_c} \sum_t \text{Securitisation/GDP}_{c,t}, \quad (6)$$

where T_c denotes the number of years available for country c . Using a time-series average reduces the influence of short-term fluctuations and provides a more stable indicator of the structural importance of securitisation in each country.

Countries are then classified using a dummy variable HighSec_c , which equals one if $\overline{\text{SecDepth}}_c$ exceeds the euro-area median, and zero otherwise.¹³ This mirrors common practice in the finance-and-growth literature, where long-run financial development indicators capture structural cross-country differences (Levine, 2005; Rajan & Zingales, 1998).

¹³See Appendix Table A5 for country classification.

The interaction specification is:

$$\begin{aligned} \log(\text{Total Loans}_{c,t}) = & \alpha + \beta_1 \log(\text{Securitisation/GDP}_{c,t-1}) \\ & + \beta_2 \left[\log(\text{Securitisation/GDP}_{c,t-1}) \times \text{HighSec}_c \right] \\ & + \gamma \mathbf{X}_{c,t-1} + \alpha_c + \lambda_t + \varepsilon_{c,t}. \end{aligned} \quad (7)$$

Here, β_1 represents the association in structurally low-securitisation countries, while β_2 captures the differential association in high-securitisation environments.

Table 5: Country-level analysis

	(1)	(2)
$\log(\text{Securitisation/GDP})$	0.094 (0.363)	-0.108 (0.242)
GDP growth	-0.015 (0.114)	-0.011 (0.256)
Unemployment rate	-0.013 (0.694)	-0.027 (0.346)
$\log(\text{Securitisation/GDP}) \times \text{HighSec}_c$		0.483* (0.017)
Observations	133	133
R^2	0.991	0.993
Within R^2	0.071	0.335
Std. errors	by: country	by: country
Country FE	Yes	Yes
Year FE	Yes	Yes

Notes: The dependent variable is the logarithm of total bank lending aggregated at the country level. Column (2) includes an interaction between lagged $\log(\text{Securitisation/GDP})$ and a dummy for high-securitisation countries. Countries are classified as high-securitisation if their average Securitisation/GDP over the sample exceeds the euro-area median. GDP growth and unemployment are lagged by one year. All regressions include country and year fixed effects. Standard errors are clustered at the country level. Numbers in parentheses report p -values. Statistical significance is denoted by *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, . $p < 0.1$.

Column (1) shows that, when pooling all euro-area countries, the coefficient on $\log(\text{Securitisation/GDP})$ is positive but statistically insignificant. This suggests that within-country changes in securitisation depth are not systematically associated with aggregate lending once fixed effects are included.

Column (2) reveals substantial heterogeneity. The coefficient on $\log(\text{Securitisation}/\text{GDP})$ is negative but statistically insignificant, capturing the association in structurally low-securitisation countries. The interaction term is positive and statistically significant (0.483, $p = 0.017$), indicating that the securitisation–lending association is stronger in countries with structurally deeper securitisation markets.

The implied elasticity in high-securitisation countries is 0.375 ($-0.108 + 0.483$). Thus, in these environments, a 1% increase in $\text{Securitisation}/\text{GDP}$ is associated with approximately a 0.38% increase in aggregate lending, while the estimated effect in low-securitisation countries is close to zero.

The country-level evidence suggests that securitisation is more closely linked to lending in environments where markets are structurally deeper and more developed. One interpretation is that securitisation requires complementary institutional conditions, such as investor participation, market liquidity, and regulatory clarity, to affect credit supply meaningfully. This pattern is consistent with theories emphasising complementarities in financial development (Acemoglu & Zilibotti, 1997; Levine, 2005) and with evidence that institutional quality shapes the effectiveness of financial deepening (Rajan & Zingales, 1998).

Importantly, these findings complement the bank-level results. While the micro-level regressions indicate a positive average association, the country-level analysis suggests that this relationship is concentrated in structurally high-securitisation environments. This nuance is relevant from a policy perspective: expanding securitisation markets may support lending primarily where supporting institutional and market infrastructures are already in place.

6. Discussion and Conclusion

The analysis in this paper set out to examine whether the depth of securitisation markets is systematically associated with bank lending in the post-crisis euro area. The findings suggest that it is. In the baseline bank-level specification, a 10% increase in $\text{Securitisation}/\text{GDP}$ is associated with an increase in lending of approximately 1.1%. While modest in magnitude, this elasticity is economically meaningful in a period characterised by subdued credit growth and balance-sheet repair. The result is broadly consistent with theoretical models in which improved asset liquidity and risk transfer mechanisms relax intermediaries' effective balance-sheet constraints and support credit

provision (Gertler & Kiyotaki, 2010; Holmstrom & Tirole, 1997). It also aligns with U.S. evidence showing that access to securitisation markets weakens the link between funding conditions and loan supply (Loutskina & Strahan, 2009).

At the same time, the results indicate that securitisation does not operate as a uniform expansionary force across all types of lending. The association appears stronger for non-corporate lending, while corporate lending responds weakly and, in some specifications, negatively. Although these estimates should be interpreted with caution, the pattern is consistent with the institutional structure of securitisation markets. Retail and mortgage-type loans are more standardised and easier to pool, whereas corporate lending is typically more relationship-based and information-intensive. As argued by Boot and Thakor (2000), banks maintain a comparative advantage in relationship lending that is less easily replicated by market-based finance. In this sense, securitisation may influence not only the volume of credit, but also its allocation across borrower segments.

The country-level results provide additional evidence for this interpretation. Within-country changes in securitisation depth are not uniformly associated with aggregate lending across the euro area. Instead, the positive relationship is concentrated in countries with structurally deeper securitisation markets. This finding is consistent with theories of financial development that emphasise complementarities between market infrastructure, legal institutions, and financial deepening. (Acemoglu & Zilibotti, 1997; Levine, 2005). Where securitisation markets are supported by investor participation, legal certainty, and established issuance practices, changes in market depth appear more closely linked to aggregate lending dynamics. In countries with persistently shallow markets, securitisation activity may be too limited or too fragmented to meaningfully affect credit supply.

The negative association between capitalisation and lending is also notable. A 5 percentage point increase in the equity-to-assets ratio is associated with a sizeable decline in lending. Rather than implying that stronger capital mechanically suppresses credit supply, this result likely reflects post-crisis balance-sheet adjustment. Following the Global Financial Crisis and the tightening of capital regulation, many European banks increased capital ratios through deleveraging and asset contraction rather than through equity issuance. Similar balance-sheet dynamics have been documented in the literature on bank capital structure and adjustment (Gropp & Heider, 2009). The estimated coefficient therefore appears to capture endogenous responses to regulatory pressure rather than a structural trade-off between capital and lending.

The heterogeneity analysis provides little support for the view that securitisation primarily

benefits weakly capitalised or high-risk banks. The interaction terms with capital and risk dummies are economically small and statistically insignificant. This contrasts with pre-crisis concerns that securitisation could disproportionately enable risk-taking by constrained institutions. In the post-crisis euro-area setting, characterised by strengthened supervision and risk-retention requirements, securitisation appears to function as a broadly similar funding channel across institutions rather than as a concentrated tool of regulatory arbitrage. This interpretation is consistent with evidence that post-crisis reforms reshaped incentives in structured credit markets (Acharya et al., 2013).

From a policy perspective, the results suggest a nuanced view of securitisation as a tool to support credit supply. Securitisation is positively associated with lending, but its effectiveness depends on institutional context and market development. Initiatives such as the Simple, Transparent and Standardised (STS) framework may therefore yield heterogeneous outcomes across countries. In environments with deeper markets and established investor bases, marginal increases in securitisation depth appear more likely to coincide with lending expansion. In less developed markets, structural frictions may limit such effects.

Several limitations qualify these conclusions. The empirical strategy relies on within-bank and within-country variation over time and includes fixed effects and observable controls. However, reverse causality cannot be fully excluded. Stronger lending growth may stimulate securitisation activity through higher loan origination, even when explanatory variables are lagged. In addition, time-varying country-specific factors such as differences in regulatory implementation, financial integration, or shifts in investor demand for structured products may jointly affect securitisation depth and lending outcomes. Although fixed effects absorb time-invariant heterogeneity and common shocks, they cannot fully eliminate bias from unobserved trends.

The limited number of country clusters also calls for caution in statistical inference. While clustering at the country level is appropriate given the level of variation in the main explanatory variable, small-cluster concerns may affect precision.

For these reasons, the estimated coefficients should be interpreted as conditional associations rather than causal effects. Establishing causal identification would require plausibly exogenous shocks to securitisation markets, for example regulatory discontinuities or institutional reforms generating differential exposure across banks or countries.

Future research could build on this analysis by exploiting regulatory reforms as quasi-

natural experiments, incorporating loan-level securitisation exposure where available, or extending the framework to a broader cross-country setting. Linking securitisation dynamics to borrower-level outcomes would also help clarify how changes in financial structure affect real economic activity.

Overall, the findings suggest that securitisation continues to be positively associated with bank lending in the post-crisis euro area, but that its effects are shaped by institutional development and loan composition. Rather than acting as a uniform engine of credit expansion, securitisation appears to operate as a liquidity and funding mechanism whose impact depends critically on market structure and regulatory context. These findings highlight the importance of institutional context in shaping the relationship between financial innovation and credit supply.

A. Appendix

Table A1: Variable Definitions and Data Sources

Variable Name	Definition	Data Source
Log Total Loans	Log of total loans, constructed as the sum of mortgage, consumer, corporate, and other loans.	BankFocus
Log Corporate Loans	Log of corporate loans, capturing lending to non-financial firms.	BankFocus
Log Non-Corporate Loans	Log of non-corporate loans, defined as total loans minus corporate loans.	BankFocus
Log Assets	Lagged log of total assets; proxy for bank size.	BankFocus
Equity/Assets	Lagged equity-to-assets ratio (winsorised at the 1st and 99th percentiles); proxy for capital strength.	BankFocus
Tier 1/RWA	Lagged Tier 1 capital divided by risk-weighted assets (winsorised); regulatory capital measure.	BankFocus
Deposit/Assets	Lagged deposits-to-assets ratio (winsorised); proxy for funding structure.	BankFocus
NPL Ratio	Lagged non-performing loans divided by total loans (winsorised); proxy for asset quality.	BankFocus
ROA	Lagged return on assets (winsorised), defined as net income over total assets.	BankFocus
Securitisation/GDP	Outstanding securitisation stock divided by nominal GDP at the country–year level.	ECB FVC; Eurostat
Log Securitisation/GDP	Lagged logarithm of Securitisation/GDP; main explanatory variable.	ECB FVC; Eurostat
GDP Growth	Lagged annual real GDP growth rate (percent).	Eurostat
Unemployment Rate	Lagged unemployment rate (age 15–74, percent).	Eurostat
Weak Capital	Dummy equal to 1 if a bank’s lagged Equity/Assets ratio is below the within-year median.	BankFocus
High Risk	Dummy equal to 1 if a bank’s lagged NPL ratio is above the within-year median.	BankFocus
High Sec	Dummy equal to 1 for countries with above-median average Securitisation/GDP over the sample period.	ECB FVC; Eurostat

Table A2: Sample Construction and Attrition

Steps	Obs.
Raw BankFocus extract	17,887
After cleaning and variable conversions	17,887
After deduplication to one bank-year observation	13,785
Euro-area banks, 2009–2022	8,662
Asset threshold and minimum reporting requirement	8,292
Final merged regression sample	8,292

Notes: The table reports the sequential sample restrictions applied to construct the final bank-year panel. The asset threshold requires total assets of at least EUR 10 million. Banks must report total loans in at least three years. The final sample includes euro-area banks matched with country-level securitisation and macroeconomic data.

Table A3: Coverage of Bank-level Securitisation Variables in BankFocus Data

Variable	Non-missing	Missing	Missing share (%)
Other exposure to securitisations	0	8,662	100.0
Managed securitised assets	10	8,652	99.9

Notes: This table reports the availability of bank-level securitisation variables in the cleaned euro-area bank-year panel ($N = 8,662$). Missing values indicate that the corresponding item is not reported in BankFocus for that bank-year. The near-complete absence of these variables prevents constructing a representative panel of bank-level securitisation exposures.

Table A4: Summary Statistics (Country–Year Panel)

	Mean	Median	P25	P75	SD	N
log(Total Loans)	26.234	26.581	23.972	28.376	2.171	252
log(Securitisation/GDP)	-1.813	-1.947	-2.452	-0.858	1.627	133
GDP Growth	1.840	1.656	0.238	2.942	4.473	138
Unemployment Rate	8.854	7.300	5.600	10.525	4.713	138

Table A5: Country Classification by Average Securitisation Depth

Country	Avg. Securitisation Depth	Years	High Securitisation
Luxembourg	3.02	14	1
Ireland	1.39	14	1
Netherlands	0.42	14	1
Spain	0.23	14	1
Belgium	0.17	14	1
Italy	0.12	14	0
Portugal	0.12	14	0
France	0.09	14	0
Malta	0.05	14	0
Germany	0.02	14	0
Austria	0.01	14	0
Cyprus	–	14	–
Estonia	–	14	–
Finland	–	14	–
Lithuania	–	14	–
Latvia	–	14	–
Slovenia	–	14	–
Slovakia	–	14	–

Notes: Average securitisation depth is defined as the mean outstanding securitisation stock-to-GDP ratio over the sample period (2009–2022). Countries are classified as high-securitisation countries if their average securitisation depth exceeds the median across Eurozone countries.

Table A6: Robustness: Regulatory Capital (Tier 1) Specification

	(1)	(2)	(3)	(4)	(5)
Log Securitisation Depth _{<i>t</i>-1}	0.231* (0.023)	0.054 (0.371)	0.176* (0.034)	0.165*** (< 0.001)	0.153** (0.003)
Log Assets _{<i>t</i>-1}		1.001*** (< 0.001)	1.015*** (< 0.001)	0.658*** (< 0.001)	0.668*** (< 0.001)
Tier1/RWA _{<i>t</i>-1}			-0.425 (0.648)	0.001 (0.999)	0.032 (0.968)
Deposit/Assets _{<i>t</i>-1}			0.920** (0.003)	0.261 (0.468)	0.206 (0.536)
NPL Ratio _{<i>t</i>-1}				0.606 (0.208)	0.607 (0.210)
ROA _{<i>t</i>-1}				2.288 (0.367)	2.162 (0.396)
Loans Components Present					0.111** (0.007)
Num. Obs.	6,192	6,192	3,309	1,029	1,029
R^2	0.952	0.967	0.972	0.981	0.981
Within R^2	0.008	0.329	0.398	0.134	0.149
Std. Errors		Clustered by country			
Bank FE	X	X	X	X	X
Year FE	X	X	X	X	X

Notes: The dependent variable is the logarithm of total loans. The main explanatory variable is the lagged logarithm of securitisation stock relative to GDP (securitisation depth). All control variables are lagged by one period and winsorised at the 1st and 99th percentiles where applicable. All regressions include bank and year fixed effects. Standard errors are clustered at the country level. Numbers in parentheses report p -values. Statistical significance is indicated by *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Table A7: Robustness: Loan Coverage ≥ 2

	(1)	(2)	(3)	(4)	(5)
Log Stock/GDP	0.259* (0.030)	0.067 (0.401)	0.090 (0.251)	0.120* (0.019)	0.134** (0.007)
Log Assets		1.021*** (;0.001)	0.949*** (;0.001)	0.730*** (;0.001)	0.716*** (;0.001)
Equity/Assets			-0.566 (0.767)	-4.174** (0.002)	-4.233** (0.003)
Deposit/Assets			1.024*** (;0.001)	0.486 (0.145)	0.426 (0.170)
NPL Ratio				0.500 (0.367)	0.576 (0.300)
ROA				2.492 (0.278)	2.466 (0.258)
Loans components present					0.133 (0.139)
Num. Obs.	4428	4428	4327	1242	1242
R^2	0.957	0.973	0.974	0.983	0.983
Adj. R^2	0.951	0.969	0.971	0.978	0.979
Within R^2	0.014	0.380	0.386	0.339	0.350
Adj. Within R^2	0.014	0.380	0.385	0.335	0.345
Std. Errors	by: country	by: country	by: country	by: country	by: country
FE: bvd_id_number	X	X	X	X	X
FE: year	X	X	X	X	X

Notes: This table reports fixed-effects panel regressions using a restricted sample of bank-year observations with at least two loan components reported. The dependent variable is the logarithm of total bank loans. All specifications include bank fixed effects and year fixed effects. Control variables are lagged by one year. Standard errors are clustered at the country level. Numbers in parentheses report p-values. Statistical significance is indicated by *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A8: Heterogeneity Across Banks: Capital Weakness and Credit Risk

	(1) Weak Capital (Tercile)	(2) High NPL (Tercile)
Log Securitisation/GDP	0.123** (0.009)	0.074+ (0.093)
Weak Capital (T3)	-0.119+ (0.095)	
Log Assets	0.785*** (< 0.001)	0.759*** (< 0.001)
Equity/Assets	-4.748** (0.003)	-4.625** (0.002)
Deposit/Assets	0.233 (0.484)	0.286 (0.355)
NPL Ratio	0.265 (0.583)	0.218 (0.632)
ROA	3.985 (0.219)	4.360 (0.142)
Loans Components Present	0.131** (0.002)	0.131** (0.001)
Log Securitisation/GDP $\times$ Weak Capital (T3)	-0.024 (0.287)	
High NPL (T3)		0.171 (0.270)
Log Securitisation/GDP $\times$ High NPL (T3)		0.076 (0.313)
Num. Obs.	1,349	1,349
R^2	0.983	0.983
Within R^2	0.378	0.379
Std. Errors	by: country	by: country
Bank FE	X	X
Year FE	X	X

Notes: The dependent variable is the logarithm of total loans. Weak Capital (T3) equals one if the bank's lagged equity-to-assets ratio lies in the lowest tercile within the year. High NPL (T3) equals one if the bank's lagged NPL ratio lies in the highest tercile within the year. All regressions include bank and year fixed effects. Standard errors are clustered at the country level. Numbers in parentheses report p -values.

+ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

References

- Acemoglu, D., & Zilibotti, F. (1997). Was Prometheus unbound by chance? risk, diversification, and growth. *Journal of Political Economy*, 105(4), 709–751.
- Acharya, V. V., & Richardson, M. (Eds.). (2009). *Restoring financial stability: How to repair a failed system*. John Wiley & Sons.
- Acharya, V. V., Schnabl, P., & Suarez, G. (2013). Securitization without risk transfer. *Journal of Financial Economics*, 107(3), 515–536. <https://doi.org/https://doi.org/10.1016/j.jfineco.2012.09.004>
- Altunbas, Y., Gambacorta, L., & Marques-Ibanez, D. (2009). Securitisation and the bank lending channel. *European Economic Review*, 53(8), 996–1009. <https://doi.org/https://doi.org/10.1016/j.euroecorev.2009.03.004>
- Basel Committee on Banking Supervision. (2011). *Basel iii: A global regulatory framework for more resilient banks and banking systems* (tech. rep.). Bank for International Settlements.
- Basel Committee on Banking Supervision. (2013). *Basel iii: The liquidity coverage ratio and liquidity risk monitoring tools* (tech. rep.). Bank for International Settlements.
- Benetton, M., Compiani, G., & Morse, A. (2021). When financial markets signal distress: The role of securitization. *Review of Financial Studies*, 34(6), 2771–2810.
- Bernanke, B. S., & Gertler, M. (1995). Inside the black box: The credit channel of monetary policy transmission. *Journal of Economic perspectives*, 9(4), 27–48.
- Bertrand, M., Duflo, E., & Mullainathan, S. (2004). How much should we trust differences-in-differences estimates? *Quarterly Journal of Economics*, 119(1), 249–275.
- Boot, A. W., & Thakor, A. V. (2000). Can relationship banking survive competition? *Journal of Finance*, 55(2), 679–713.
- Brunnermeier, M. K., & Oehmke, M. (2013). The maturity rat race. *Journal of Finance*, 68(2), 483–521.
- Casu, B., Clare, A., Sarkisyan, A., & Thomas, S. (2013). Securitization and bank performance. *Journal of Money, Credit and Banking*, 45(8), 1617–1658.
- Duffie, D., & Gârleanu, N. (2001). Risk and valuation of collateralized debt obligations. *Financial Analysts Journal*, 57(1), 41–59.
- European Central Bank. (2018). Securitisation in the euro area. <https://www.ecb.europa.eu/pub/pdf/scpops/ecb.op201.en.pdf>
- European Commission. (2015). Action plan on building a capital markets union. <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:52015DC0468>

- European Union. (2017). Regulation (eu) 2017/2402 of the european parliament and of the council.
- Gennaioli, N., Shleifer, A., & Vishny, R. (2012). Neglected risks, financial innovation, and financial fragility. *Journal of Financial Economics*, 104(3), 452–468.
- Gertler, M., & Kiyotaki, N. (2010). Financial intermediation and credit policy in business cycle analysis. In B. Friedman & M. Woodford (Eds.), *Handbook of monetary economics* (pp. 547–599, Vol. 3). Elsevier.
- Gorton, G., & Metrick, A. (2012). Securitized banking and the run on repo. *Journal of Financial Economics*, 104(3), 425–451.
- Gorton, G. B., & Pennacchi, G. G. (1995). Banks and loan sales marketing nonmarketable assets. *Journal of Monetary Economics*, 35(3), 389–411. [https://doi.org/https://doi.org/10.1016/0304-3932\(95\)01199-X](https://doi.org/https://doi.org/10.1016/0304-3932(95)01199-X)
- Gropp, R., & Heider, F. (2009). The determinants of bank capital structure. *Review of Finance*, 14(4), 587–622.
- Gropp, R., Mosk, T., Ongena, S., & Wix, C. (2019). Banks' response to higher capital requirements. *Review of Finance*, 23(3), 645–676.
- Heider, F., Saidi, F., & Schepens, G. (2019). Life below zero: Bank lending under negative policy rates. *The Review of Financial Studies*, 32(10), 3728–3761. <https://doi.org/10.1093/rfs/hhz016>
- Holmstrom, B., & Tirole, J. (1997). Financial intermediation, loanable funds, and the real sector. *Quarterly Journal of Economics*, 112(3), 663–691.
- Jiménez, G., Ongena, S., Peydró, J.-L., & Saurina, J. (2012). Credit supply and monetary policy. *American Economic Review*, 102(5), 2301–2326.
- Jiménez, G., Ongena, S., Peydró, J.-L., & Saurina, J. (2014). Hazardous times for monetary policy. *Review of Economic Studies*, 81(2), 463–505.
- Jiménez, G., Ongena, S., Peydró, J.-L., & Saurina, J. (2017). Macroprudential policy, countercyclical bank capital buffers, and credit supply: Evidence from the spanish dynamic provisioning experiment. *Journal of Political Economy*, 125(6), 2126–2177.
- Jones, D. (2000). Emerging problems with the basel capital accord: Regulatory capital arbitrage and related issues. *Journal of Banking & Finance*, 24(1-2), 35–58.
- Kashyap, A. K., Rajan, R. G., & Stein, J. C. (2002). Banks as liquidity providers: An explanation for the coexistence of lending and deposit-taking. *The Journal of Finance*, 57(1), 33–73.
- Kashyap, A. K., & Stein, J. C. (2000). What do a million observations on banks say about monetary policy transmission? *American Economic Review*, 90(3), 407–428.

- Keys, B. J., Mukherjee, T., Seru, A., & Vig, V. (2010). Did securitization lead to lax screening? evidence from subprime loans. *Quarterly Journal of Economics*, 125(1), 307–362. <https://doi.org/10.1162/qjec.2010.125.1.307>
- Khwaja, A. I., & Mian, A. (2008). Tracing the impact of bank liquidity shocks. *Quarterly Journal of Economics*, 123(4), 1413–1461.
- Levine, R. (2005). Finance and growth: Theory and evidence. *Handbook of Economic Growth*, 1, 865–934.
- Loutskina, E. (2011). The role of securitization in bank liquidity and funding management. *Journal of Financial Economics*, 100(3), 663–684.
- Loutskina, E., & Strahan, P. E. (2009). Securitization and the declining impact of bank finance on loan supply. *Journal of Finance*, 64(2), 861–889.
- Mian, A., & Sufi, A. (2009). The consequences of mortgage credit expansion: Evidence from the u.s. mortgage default crisis. *Quarterly Journal of Economics*, 124(4), 1449–1496. <https://doi.org/10.1162/qjec.2009.124.4.1449>
- Peek, J., & Rosengren, E. (2000). Collateral damage. *American Economic Review*, 90(1), 30–45.
- Pennacchi, G. G. (1988). Loan sales and the cost of bank capital. *The Journal of Finance*, 43(2), 375–396.
- Rajan, R. G., & Zingales, L. (1998). Financial dependence and growth. *American Economic Review*, 88(3), 559–586.

B. Declaration of Academic Integrity

1. I hereby confirm that this work – or in case of group work, the contribution for which I am responsible and which I have clearly identified as such - is my own work and that I have not used any sources or resources other than those referenced.

I take responsibility for the quality of this text and its content and have ensured that all information and arguments provided are substantiated with or supported by appropriate academic sources. I have clearly identified and fully referenced any material such as text passages, thoughts, concepts or graphics that I have directly or indirectly copied from the work of others or my own previous work. Except where stated otherwise by reference or acknowledgement, the work presented is my own in terms of copyright.

2. I understand that this declaration also applies to generative AI tools which cannot be cited (hereinafter referred to as ‘generative AI’).

I understand that the use of generative AI is not permitted unless the examiner has explicitly authorized its use (Declaration of Permitted Resources). Where the use of generative AI was permitted, I confirm that I have only used it as a resource and that this work is largely my own original work. I take full responsibility for any AI-generated content I included in my work.

Where the use of generative AI was permitted to compose this work, I have acknowledged its use in a separate appendix. This appendix includes information about which AI tool was used or a detailed description of how it was used in accordance with the requirements specified in the examiner’s Declaration of Permitted Resources.

I have read and understood the requirements contained therein and any use of generative AI in this work has been acknowledged accordingly (e.g. type, purpose and scope as well as specific instructions on how to acknowledge its use).

3. I also confirm that this work has not been previously submitted in an identical or similar form to any other examination authority in Germany or abroad, and that it has not been previously published in German or any other language.

4. I am aware that any failure to observe the aforementioned points may lead to the imposition of penalties in accordance with the relevant examination regulations. In particular, this may include that my work will be classified as deception and marked as failed. Repeated or severe attempts to deceive may also lead to a temporary or permanent exclusion from further assessments in my degree programme.

Berlin, 15.03.2026

Place and date

Signature